

Customer Churn Prediction

A supervised machine learning pipeline for binary churn classification using Logistic Regression and Random Forest across two datasets.

Presented to: Professor Luigi Capone

Tools: Python – PySpark (Spark MLlib) - pandas - matplotlib

Models evaluated: Logistic Regression - Random Forest

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Dataset Overview, Preprocessing Pipeline & Model Selection

Datasets

Dataset 1

Training set: 31,922 rows
Test set: 8,078 rows
51.6% churn rate

Dataset 2

Training set: 39,902 rows
Test set: 10,098 rows
87,2% churn rate

Preprocessing Pipeline

1

80 / 20 Train-Test Split

Random split via `DataFrame.randomSplit([0.8, 0.2], seed=23)` for reproducibility. Stratification preserves churn ratio in both sets.

2

Column-type Detection

Numerical vs. categorical columns identified from the Spark DataFrame schema (dtypes).

3

Numerical Pipeline

Median imputation (Imputer, strategy='median') → StandardScaler normalisation.

4

Categorical Pipeline

Mode imputation → StringIndexer + OneHotEncoder (handleInvalid='keep').

Model Architecture & Selection

Two classifiers were trained end-to-end inside a single Spark ML Pipeline(`pyspark.ml.Pipeline`) chaining preprocessing stages and the estimator.. The best model per dataset was selected by ROC AUC on the held-out test set.

Random Forest

Key parameters

- `n_trees`: default (20)
- `class_weight`: 'balanced'
- `random_state`: 420

BASELINE

Captures non-linear feature interactions; robust to outliers; no scaling required. Used as a strong baseline.

ROC AUC — DS1: 0.873 | DS2: 0.950

Logistic Regression

Key parameters

- `max_iter`: 1000
- `class_weight`: 'balanced'
- `random_state`: 420

SELECTED

Linear classifier with probabilistic output. Coefficients are directly interpretable as feature-level influence on churn probability.

ROC AUC — DS1: 0.901 | DS2: 0.969

Choosing the Right Model for a Churn Problem

The choice of model directly determines what actions the business can take, who gets targeted, and how resources are allocated. For this reason, is not only a statistical modelling decision.

The Two Metrics That Drive Retention Decisions



Precision
98.8%

*Of all customers we flag as churners,
how many actually churn?*

Low precision = wasted retention spend. The business pay for campaigns targeting loyal customers who were never going to leave.



Recall
89.1%

*Of all customers who actually churn,
how many did we catch?*

Low recall = missed revenue. Real churners slip through without ever receiving a retention offer.

Model Choice Through a Marketing Lens

Interpretability

RF *Black-box. You know who to target, but not why.*

LR Each feature has a coefficient; in this way the firm know exactly which variable (e.g. low loyalty score) is pushing a customer toward churn. Actionable for campaign design.

False Positive Cost

RF *Slightly more false positives in this dataset → more wasted outreach budget.*

LR Precision of 98.9% → minimal budget wasted on customers who are not at risk.

Probability Output

RF *Probability scores available but less calibrated for tabular data.*

LR Well-calibrated churn probability score → enables tiered campaigns (e.g. high-risk vs. medium-risk segments).

Model Performance Results

Dataset 1 - 51.6% churn rate

Best model: Logistic Regression



Confusion Matrix

	Pred: 0	Pred: 1
Real: 0	3,171	676
Real: 1	818	3,422

Balanced performance across both classes.
AUC of 0.901 indicates strong discriminative ability.

Dataset 2 - 87,2% churn rate

Best model: Logistic Regression



Confusion Matrix

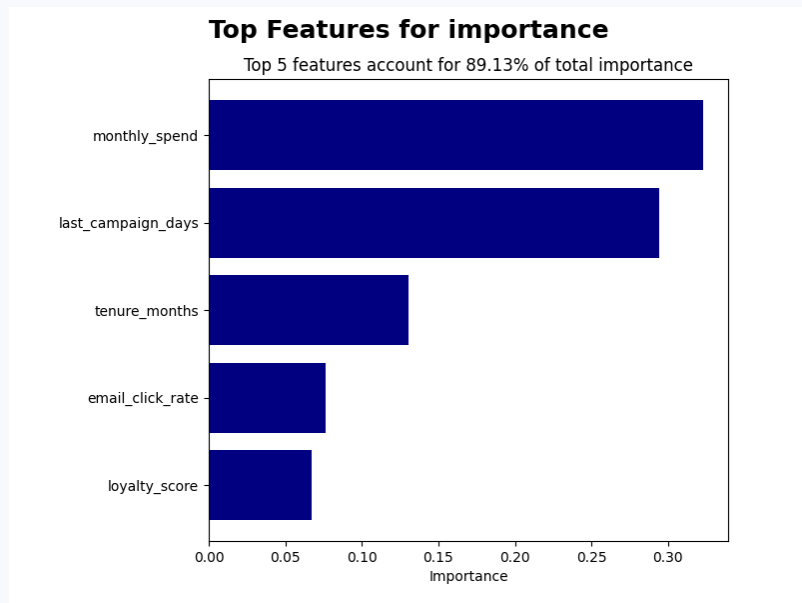
	Pred: 0	Pred: 1
Real: 0	1,140	99
Real: 1	935	7,924

Higher churn base rate yields a stronger signal.
Precision of 98.8%, very few false positives.

Interpretation & Key Takeaways

Top Predictive Features

Logistic Regression coefficient magnitudes (normalised, Dataset 1)



The top 5 features account for most of the predictive weight.

Conclusions

Logistic Regression outperformed Random Forest

Despite being a simpler model, it achieved higher ROC AUC on both datasets, confirming that a well-preprocessed linear model can be very competitive for tabular churn data.

Pipeline design prevented data leakage

By fitting all transformers exclusively on training data and applying them to the test set, the evaluation metrics genuinely reflect out-of-sample performance.

Class imbalance was addressed systematically

The `class_weight='balanced'` parameter ensured neither class dominated the loss function, particularly important for Dataset 2 with its 87% churn majority.

Thank you

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